PREDICTING SECOND-HAND VEHICLE PRICES IN KENYA USING DATA-DRIVEN MACHINE LEARNING TECHNIQUES

BY:

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UNITED STATES INTERNATION UNIVERSITY - AFRICA

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A Research Proposal Submitted to School of Science and Technology in Partial Fulfillment of Requirements for the Award of Degree of Bachelor of Science in Data Science and Analytics

UNITED STATES INTERNATIONAL UNIVERSITY- AFRICA

STUDENT'S DECLARATION

I, the undersigned, declare that this research is my original work, except for cited quotations and references that has properly been acknowledged and referenced in accordance with the USIU-Africa requirements. Additionally, this research has not been submitted to any other college, institution, or any other university for examination, award of degree or publication or academic credit.

Signed ..... **Date** .....

Daki Lelo Dima (668925)

SUPERVISOR APPROVAL

This research proposal has been submitted for examination with my approval as the University Supervisor

Signed ..... **Date** .....

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ABSTRACT

This study develops a machine learning-based model for predicting second-hand vehicle prices in Kenya using publicly available asking-price listings obtained from Jiji.co.ke and Cheki Kenya. The Kenyan used car market is characterized by information asymmetry, informal pricing practices, and the absence of standardized valuation mechanisms, creating difficulties for buyers, sellers, and financial institutions in determining fair vehicle prices. The study examines the relationship between vehicle attributes and price and compares the predictive performance of Multiple Linear Regression, Random Forest, and XGBoost models. A quantitative cross-sectional design is employed, with structured data preprocessing, Kenya-specific feature engineering, stratified five-fold cross-validation, and model evaluation using Mean Absolute Percentage Error and Root Mean Squared Error. Model interpretability is enhanced through SHAP values, while disaggregated performance analysis is used to assess predictive fairness across common and rare vehicle segments. The findings are expected to support the development of a deployable vehicle price prediction tool that improves market transparency, informs lending decisions, and contributes to the formalization of Kenya's automotive sector**.**

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Special appreciation is extended to the platforms and institutions that made publicly available data accessible for this research. I also acknowledge the support of my family for their patience, understanding, and unwavering encouragement throughout this academic journey.

DEDICATION

This work is dedicated to my family for their love, support, and sacrifices, which have made my academic journey possible.

It is also dedicated to scholars and practitioners working toward data-driven solutions for African markets, and to all individuals committed to promoting transparency, fairness, and innovation in the automotive sector.

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LIST OF ABBREVIATIONS

AUC	Area Under the Curve
CBK	Central Bank of Kenya
GDP	Gross Domestic Product
IMF	International Monetary Fund
IRB	Institutional Review Board
KES	Kenyan Shillings
KIPPRA	Kenya Institute for Public Policy Research and Analysis
KNN	K-Nearest Neighbors
KNBS	Kenya National Bureau of Statistics
KRA	Kenya Revenue Authority
MAPE	Mean Absolute Percentage Error
MLR	Multiple Linear Regression
NACOSTI	National Commission for Science, Technology and Innovation
RMSE	Root Mean Squared Error
SHAP	Shapley Additive exPlanations
SMOTE	Synthetic Minority Oversampling Technique
SMOTE-NC	Synthetic Minority Oversampling Technique for Nominal and Continuous features

SVM	Support Vector Machine
UAE	United Arab Emirates
UK	United Kingdom
USD	United States Dollar
USITC	United States International Trade Commission
XGBoost	Extreme Gradient Boosting

CHAPTER ONE: INTRODUCTION

1.1 Introduction

The motor vehicle sector of Kenya is dominated by second-hand vehicles that are brought from other countries, where nearly 85% of all motor vehicle purchases are made of imports that were previously owned by people in Japan, the United Kingdom, and the United Arab Emirates (Kenya National Bureau of Statistics [KNBS], 2023). This high number happens because family income is small, the production of motor vehicles within the country is limited, and the rules for importing focus on giving affordable mobility to individuals with low income (Mordor Intelligence, 2026). Experts claim that the Kenyan used car market was valued at approximately USD 1.28 billion in 2025 and is projected to reach USD 1.54 billion by 2031, growing at a CAGR of 3.17% (Mordor Intelligence, 2026). Because many people believe that the demand is staying high, KNBS (2023) showed that 78,127 motor vehicles arrived in 2023, and Japan provided 94% of these used cars.

Hiding price information is a big problem because the buyer and the seller do not see the same facts. In the markets that are not official, like unregistered dealerships or websites where people talk to each other, a standard price for used cars is not found (Shibia, 2022). Many people believe that the lack of clear pricing hurts the market. Kenya does not have a system to decide value that looks at the age of the motor vehicle or how the bad roads of Kenya make parts wear out, even though Japanese brands have their own special costs for maintenance (U.S. International Trade Commission, 2020). It has been observed that machine learning, especially methods like XGBoost and Random Forest, can predict the price of used cars well by looking at how different features work together. Onserio and Mukudi (2025) represent a very similar Kenyan study which reported that a local stacking model was very accurate. Because internet platforms like Jiji.co.ke and Cheki

Kenya are growing and dealers are using more tools for their stock, the possibility for a strong model made for Kenya has improved (Mwangi, 2023).

1.2 Problem Statement

The market for used cars in Kenya does not have prices that are clear or based on data, and this lack of clarity leads to mistakes in value that hurt the people (Shibia, 2022). Price is usually decided by talking and arguing between people because the models from Western countries do not fit the way motor vehicles lose value in Kenya (Mordor Intelligence, 2026). Experts claim that three barriers keep this problem alive. First, the location matters because Dutulescu et al. (2023) showed that a model made for one place must be changed a lot to work in a different place. Second, the way we calculate prices fails when a motor vehicle is a type that people rarely see, even when using neural networks (Zheng, 2025). Third, data from the informal sellers is not recorded by KNBS, so the price movements in those areas are not known (Shibia, 2022). It has been observed that without a framework for Kenya, the (Republic of Kenya 2022) will not reach its goals.

1.3 Objectives

1.3.1 General Objective

The purpose of this work is to create an accurate machine learning model that calculates the fair price for used cars in Kenya using local data.

1.3.2 Specific Objectives

1. To find the features of a motor vehicle that change the price most in Kenya by using correlation analysis and Random Forest scores.

2. To train and compare three models; Linear Regression, Random Forest, and XGBoost so that the best one is chosen based on error scores like MAPE and RMSE.

3. To check if the model works for used cars that are common and used rare cars by using special counting methods and oversampling.

1.4 Research Questions

Which features of a motor vehicle, such as the miles driven, the years of age, and the county where it stays, have the biggest power over the price of used cars in Kenya?

Which machine learning way gives the price that is most correct for the used cars of Kenya?

Does the model give a price that is just as good for motor vehicle brands that are not common?

1.5 Justification

This study looks at a problem that changes the lives of many people because correct prices help individuals with low income when they buy a motor vehicle (Mordor Intelligence, 2026). Banks will also find this work useful when they need to know the value of used cars that are used as security for loans, which is important because many loans in Kenya are not being paid back (Central Bank of Kenya, 2023). Many people believe that this study helps the Consumer Protection Act (2012, s. 3(4)(d)–(e)) and the National Automotive Policy (Republic of Kenya, 2022). This research builds a special way to look at data that includes the county demand and the history of how the motor vehicle was brought into the country (Dutulescu et al., 2023). It has been observed

that by using SMOTE-NC and other special methods, the used cars that are rare can also be priced correctly (Zheng, 2025).

1.6 Scope

This work uses only the prices that are shown on Jiji.co.ke and Cheki Kenya, which are the two main online markets for motor vehicles in Kenya. Data from the Kenya Revenue Authority or the actual papers from dealers are not included. Because most people list their motor vehicles in Nairobi, the study looks mostly at that city, though Mombasa and Kisumu are also included where there is enough data. Rural areas are not part of this work because not many motor vehicles are listed there on the internet. Experts claim that because the data comes from asking prices, the real selling price might be a bit different. This study only looks at used cars for people and small trucks, so very expensive motor vehicles over KES 10 million are left out. The way the data is analyzed uses Multiple Linear Regression, Random Forest, and XGBoost with five-fold cross-validation. It has been observed that the study does not look at the words in the ads or the mechanical reliability of the motor vehicle.

1.7 Expected Contributions

The study brings help to two different areas. In a practical way, it gives a tool to find the price of used cars, which helps the banks and the buyers in the digital markets of Kenya. Methodologically, it creates a new way to engineer features that look at Kenyan roads and the rules for the age of a motor vehicle. This study also makes a way to check if the model is correct for the rare motor vehicle brands by using separate MAPE and RMSE scores. Many people believe that this research supports the National Automotive Policy and helps the people under the Consumer Protection Act (2012, s. 3(4)(d)–(e)). It has been observed that this work creates a way for other countries in Africa to build their own tools for pricing motor vehicles.

1.8 Chapter Summary

This chapter has built the start for a machine learning model to find the price f used cars in Kenya. The text explained the social situation and the trouble that comes when price information is hidden between buyers and sellers. Three main problems were shown: the bias of location, the trouble with rare segments, and the gap between the asking price and the real price. The objectives and the research questions were written to fix these gaps. It has been observed that the study uses data from Jiji.co.ke and Cheki Kenya to reach its goals. Because the study uses special feature engineering and looks at rare brands, many people believe that it will help the automotive sector of Kenya grow

CHAPTER TWO: LITERATURE REVIEW

2.1 Introduction

The current segment offers an organized examination of academic writings that relate to the act of creating a computational tool for estimating the cost of previously owned motor vehicles within the Kenyan environment. It has been observed that the used car market in Kenya is defined by a persistent lack of shared information, informal ways of trading, and the absence of standard rules for valuing things. These conditions make research on data-driven pricing both difficult for the mind and of great importance for daily life. To place the present academic inquiry within existing knowledge, the review looks at progress in machine learning methods used for automotive price prediction. The review also looks at how preparing data and selecting features help in making the precision of measurement better. Strategies for handling the problem of rare-segment prediction in long-tailed automotive datasets are included because these challenges are common in the industry. It is believed by experts that the role of interpretability in models is very high when the money involved is a large amount.

2.2 General Literature Review

2.2.1 Machine Learning in Regression and Price Prediction Tasks

Guided computational learning has grown to be the primary method for price prediction tasks inside automotive markets where ensemble ways consistently beat single-estimator ways. XGBoost was documented by Zheng (2025) to be a tool that performs better than basic linear regression and deep neural networks when looking at organized data of cars. Experts claim that the strength of XGBoost comes from the capacity of the model to handle missing values and complex interactions between features. When Huang (2025) used XGBoost on AutoTrader data, the color and body shape of the car were found to have measurable effects on the cost along with the distance traveled and age. Right-skewed residuals were also documented in that study, which shows that the problem of rare-segment prediction must be handled with care. Random Forest, which is a method based on grouping many trees, has also shown high strength in many tests. It has been observed that Random Forest is the most trustworthy tool across different automotive datasets (Selvarathi, Bhava Dharani, & Pavithra, 2023). This finding matches the work of Onserio and Mukudi (2025), who reached an accuracy level of 97.25% in Kenya when mileage, vehicle age, engine size, and usage type were used as the main factors for calculating the cost. Even though Random Forest is stronger, Multiple Linear Regression (MLR) is still utilized by the Kenya National Bureau of Statistics [KNBS] (2023) because the results are easy for humans to understand. Nandan and Ghosh (2023) found that Random Forest made the error smaller by 30% compared to MLR while keeping the order of feature importance the same. Because buyers often lack knowledge about how to value a vehicle, regression models that give a specific number

instead of a range are very helpful in reducing the gap in information (Alexstan, Monesh, Poonkodi, & Raj, 2023; Shibia, 2022).

2.2.2 Feature Engineering for Automotive Price Prediction

The act of building specific data features is recognized as a factor that decides how well a system predicts things. Hankar, Birjali, and Beni-Hssane (2022) noted that features belonging to the specific field, such as the age of the vehicle and the ratio of distance traveled per year, reduce errors more than using raw data. It has been observed that detailed equipment like special wheels or parking help can make the pricing of the vehicle more exact (Bergmann & Feuerriegel, 2025). Location is also a very important factor for the cost. Experts claim that pricing models need to be changed when they move from one country to another because the roads and what buyers like are different. In the Romanian and German markets, the value drivers were shaped by the local infrastructure and the rules for bringing in cars (Dutulescu et al., 2023). When looking at the Kenyan car market, the fact that 85% of vehicles come from overseas must be considered (Mordor Intelligence, 2026). The taxes for importing, the preference for right-hand drive, and the country where the car was made must be used as clear features. These specific details are often missing from datasets made in Western countries, but the Kenya-specific features are needed for a model to work well in Nairobi or other urban areas.

2.2.3 Model Interpretability and Explainability

The ability to explain how a model works has become a very important rule for when a vehicle is used as a guarantee for a loan. The SHAP framework is used as a way to show how each feature helps in making the final price prediction. Many people believe that SHAP rankings match what people who work in the car industry think, which makes financial organizations trust the model

more (Zheng, 2025). Nandan and Ghosh (2023) say that SHAP results should be a standard part of any tool used in growing markets. In the nation of Kenya, the Consumer Protection Act (2012, s. 3(4)(d)–(e)) requires that consumers be protected from misleading conduct and that they be provided with sufficient information to make informed choices, which includes the basis on which a price is determined. Transparency is also described as a required condition for making the car sector more formal by the National Automotive Policy (Republic of Kenya, 2022). Opaque black-box models are not allowed to be used because they cannot satisfy these legal requirements.

2.2.4 Handling Class Imbalance and Rare-Segment Prediction

A very common problem in car research is that some types of cars appear very many times while others appear very few times. Standard models are documented by Zheng (2025) to have very large errors for cars that are not common. Sometimes the error for rare cars is three times larger than the error for common cars. To fix this, ways to add more data like SMOTE or ways to make the loss function more sensitive to rare items are used. Many people believe that weighting the classes works better than just adding more data when the rare cars are very different from each other. The present academic inquiry tests both ways against a basic model to see which one works better. Error rates for the common cars and the rare cars are calculated separately to ensure the vehicle price is correct for everyone.

2.3 Empirical Review

2.3.1 Machine Learning-Based Used Car Price Prediction in Kenya

The most relevant previous works are two academic inquiries from the Kenyan environment. Onserio and Mukudi (2025) built a grouping of tools using Random Forest, SVM, and Gradient

Boosting to reach 97.25% precision for Kenyan automotive data. Mileage and car age were found to be the top factors for determining the price. In another work, Onyiego, Anyika, and Obuhuma (2025) created a regression-based model for the city of Nairobi using data from local listings. These academic inquiries work together to give a starting point for the current research. Onserio and Mukudi (2025) give a goal for precision across the whole nation, while Onyiego et al. (2025) provide a starting point for Nairobi. It has been observed that ensemble methods can capture how different features work together even when data is hard to find (Sharma et al., 2025). The gap in these Kenyan studies is that they did not look at how prices change between different counties or how rare cars behave. The present study is designed to fix these gaps.

2.3.2 Cross-Market Automotive Pricing Studies

When models were compared between Romania and Germany, it was found that they needed to be changed a lot to work in a new place (Dutulescu et al., 2023). This shows that the factors for a vehicle price depend on the place and that models from the West cannot be used in Kenya without changes. Gao (2024) looked at both MLR and Random Forest and found that Random Forest was better. The year the car was made was the most important factor, followed by distance traveled and fuel type. This order of importance is very similar to what was found in Kenya by Onserio and Mukudi (2025). It has been observed that cleaning the data and removing unusual numbers makes the model perform much better (Ather et al., 2023). Preprocessing the data is a required step because the quality of data taken from Kenyan websites is often not the same.

2.3.3 Pre-Owned Car Pricing Studies in Emerging Market Contexts

In India, Random Forest was found to be the strongest way to predict the price of a car (Nandan and Ghosh, 2023). Fuel type and the way the gears change were very important features in both

India and China (Gao, 2024). Experts claim that the importance of the brand and how much fuel a car uses changes depending on the rules of the country and how much money people earn. This is why Kenya-specific features are being built in this study. The Central Bank of Kenya (2023) has shown that many people cannot pay back their loans because the value of their vehicle was not calculated correctly. Because the used car market in Kenya is mostly Japanese imports and is located in Nairobi, the market is different from other places (KNBS, 2023). This situation makes locally trained models very necessary for the country.

2.3.4 Optimization Strategies for Rare-Segment Prediction

Zheng (2025) gives evidence that rare-segment prediction is very hard because cars that appear less than 1% of the time have very high errors. Ensemble methods that use weights for classes give the best improvement without making the model too specific to the training data. This is very important for the Kenyan market because Toyota makes up 40% of the cars (Mordor Intelligence, 2026). If a model fails for rare cars, the failure might not be seen in the total average but many people would still be affected. Separate error rates are calculated for common and rare cars in this study to prevent this problem.

2.3.5 Financial Institutions and Vehicle Valuation

The Central Bank of Kenya (2023) says that valuing a vehicle is a big challenge for managing risk. If the price of a vehicle is not known, the chance of a loan not being paid is higher. Accurate models help the financial organizations that give money by providing a benchmark that is based on data. It has been observed that the informal economy is a very big part of the wealth in Sub-Saharan Africa (Medina, Jonelis, & Cangul, 2017). For the 15 million Kenyans who work in the informal economy, a vehicle is often the only thing they own that can be used for a loan (Laterite,

2024). Because of this, the accuracy of a car price has a direct effect on whether people can get financial help.

2.4 Research Gaps

2.4.1 Geographic Bias in Existing Pricing Models

Most studies in Kenya have looked only at the market in Nairobi and thought it was the same as the whole country. However, the counties in Kenya have different roads, different amounts of money, and different needs for cars. It has been observed that no study has looked at the car price at the county level before. The present academic inquiry fixes this gap by using demand indices from Jiji.co.ke and Cheki Kenya for different counties. This allows the model to see how the place affects the price of the vehicle.

2.4.2 Rare Vehicle Segment Performance

The total accuracy of a model often hides the fact that it does not work for cars that are not common. Zheng (2025) showed that this is a serious problem, but Kenyan studies have not reported the error for rare cars yet. If the model is not tested for rare cars, the people who buy or sell those cars will not have good information. The present study fills this gap by looking at common and rare cars separately and using oversampling if the rare cars have too much error.

2.4.3 Asking-Price versus Transaction-Price Divide: A Gap for Future Research

Current studies in Kenya use data from digital listings, which show the price the seller wants, not the price that was actually paid. Many people believe that many car sales happen in informal ways without any digital record (Shibia, 2022). This means the actual price might be different from the price on the website. Models that use only website data might have a small shift in the price.

Finding the actual transaction price is a very difficult task for the future and is mentioned here as an important gap for other researchers to solve.

2.4.4 Policy-Relevant and Granular Equipment Feature Integration

Rules about importing cars and the effect of roads on the machine parts of the vehicle are not usually in the models. It has been observed that adding details about the equipment of the car makes the prediction much better in Europe (Bergmann & Feuerriegel, 2024). This study asks if things like service history and import papers can make the model better in Kenya too. This research is among the first to put these Kenya-specific features into a system for predicting the cost.

2.5 Chapter Summary

The current segment examined the academic writings that inform a machine learning-based tool for predicting the price of vehicles in the used car market of Kenya. Experts claim that ensemble ways like XGBoost and Random Forest are the best choices for looking at organized data about cars. It has been observed that the accuracy of the model becomes much higher when age, mileage, and location are used as features. Interpretability using SHAP is needed so that the model follows the laws of Kenya and can be trusted by people. The challenge of rare cars is a major problem that this study will look at very closely. The divide between the price on the website and the price paid in person is identified as a goal for future academic inquiry. The empirical review showed that Onserio and Mukudi (2025) and Onyiego et al. (2025) have already set the standard for what a model should do in Kenya. It was also found that financial organizations need these tools to help people get loans more easily. Four gaps were found: the focus only on Nairobi, the lack of data on rare cars, the divide in prices, and the lack of local policy features. The present academic inquiry

works to fix the first, second, and fourth gaps to help the field of machine learning in growing markets.

CHAPTER THREE: METHODOLOGY

3.1 Introduction

In the Kenyan land, the way to build a machine learning-based price prediction model for second-hand vehicles is shown in this chapter. This detailed academic investigation uses a numeric and predictive design that sits on supervised machine learning because it must estimate continuous vehicle prices from data that has a structure. It has been observed that the methodology answers the three specific goals from Chapter One directly. These goals look for vehicle traits that tell the price in Kenya, compare how many machine learning algorithms perform, and check if the model is fair for both common and rare vehicle types. Each section explains the reasons and the actions taken so that the work remains honest and can be copied by other people.

Limitations of earlier Kenyan works identified in the literature search give the shape to this plan, especially where geographic details and rare car data were missing (Onserio & Mukudi, 2025; Onyiego et al., 2025). This scientific exploration uses digital data from websites just like older studies did, but the boundaries are clearly stated by focusing on counties where many cars are listed. Many people believe that the gap between the price a seller asks for and the price a buyer actually pays is a hard truth of this market (Shibia, 2022). To fix the gaps in current knowledge, this work uses special feature building for Kenya and a divided way of checking the model performance.

3.2 Design/Experiment Methodology

The structure for the experiment is built like a pipe that has five stages: getting data, cleaning the details, training the machine learning-based price prediction model for second-hand vehicles,

checking how it works for rare cars, and explaining the results with SHAP values. This plan makes sure that every part connects back to the goals of the work so that it is useful for real-world use.

Three types of math rules are used for the work: Multiple Linear Regression (MLR), Random Forest, and XGBoost. MLR is used as the basic level for comparison because it matches earlier Kenyan works and gives a floor for the other methods (Kenya National Bureau of Statistics [KNBS], 2023; Nandan & Ghosh, 2023). Random Forest is picked because it is very strong for car data and performed well in prior Kenyan work (Onserio & Mukudi, 2025; Selvarathi et al., 2023). XGBoost is also used because it handles hard patterns in data very well (Zheng, 2025; Huang, 2025). Many people believe that Root Mean Squared Error (RMSE) and Mean Absolute Percentage Error (MAPE) are the best ways to see which model is king. MAPE is given more weight because it shows the error as a part of the full price, which matters most to buyers (Gao, 2024).

A special way of checking the models is also done for different car groups. It has been observed that errors can be bigger for rare cars in these types of studies (Zheng, 2025). Because of this, the data is split by how often a brand appears in the records. If the error for rare cars is three times bigger than for common cars, special math fixes like SMOTE oversampling are used to make things better.

3.3 Research Design

Scientific exploration uses a numeric and applied design by taking data from many places at one time. A supervised regression task is the way the analytical problem is solved. The price of the vehicle is the thing we want to know, and it is counted in Kenyan shillings (KES) as a flowing number. Structured vehicle traits, geographic signs, and details about car imports are used to guess this price. Classification is not used because that would hide the real price details into broad groups. Experts claim that keeping these details is important for the people who buy cars and the banks that check for risks (Alexstan et al., 2023).

Naturally occurring variations in car traits and prices on public websites are the source of knowledge for this research. No one is picked by luck and no outside change is forced because the goal is to see the links between features and realized prices (Gao, 2024). Stratified five-fold cross-validation is used so that the results stay steady and can be used for other data. This way of working is the same as the path taken by Onserio and Mukudi (2025) and follows the rules of Hankar et al. (2022) for car research.

3.4 Data Collection Methods

Gathering information for the project happens by using computer programs to look at car websites that everyone can see. No partnerships with car sellers or bank records are used for this work because it follows a specific data strategy. The fact that the data only shows the price the seller wants is clearly stated as a limit. Many experts claim that the price on a website is just a wish and not the real money paid in the end, and this reality is carried through the findings.

3.4.1 Web Scraping of Digital Listing Platforms

Computer programs collect car details from Jiji.co.ke and Cheki Kenya. These are the two biggest online selling places in the Kenyan land (Mwangi, 2023). These platforms are chosen because they have many listings and cover the whole country's digital market. A Python tool is used so that the price, name of car, year, and miles are saved for each listing. Only normal passenger cars and small work trucks are kept in the list. Motorcycles and very expensive luxury cars over KES 10 million are removed. Many people believe that Nairobi, Mombasa, and Kisumu are the main places for these sales, so listings from counties with very few cars are left out. Data from years 2022 to 2025 is used to show how the market grew after the sickness of post-COVID (Mordor Intelligence, 2026). Any cars that appear twice are removed by checking the name, year, and miles before the analysis begins.

3.5 Design/Experiment Procedures

3.5.1 Preprocessing and Feature Engineering

Clean data for the work is made by using Python tools like pandas and scikit-learn. The steps follow the high rules set by Ather et al. (2023) so that the results are good. If a car listing is missing the price, the model, or the year, that listing is thrown away. For features like engine size that have small gaps, the middle number from the same car type is used for the fill. Experts claim that using the middle number of the specific group is better than using one number for everything.

Extreme prices are fixed by using math rules so they do not pull the model too far. If a car has more than 500,000 km, it is checked by hand because it might be a typing mistake. New details like the age of the car and the miles per year are created by the researchers. The names of the cars are turned into numbers using the average price to avoid cheating the model with future knowledge (Bergmann & Feuerriegel, 2025). A log transform is applied to the price and the miles because these numbers often lean to one side in Kenya (Huang, 2025). The whole pile of data is split so that 80% is for training and 20% is for testing.

3.5.2 Model Training and Hyperparameter Optimization

Machine learning-based price prediction model for second-hand vehicles is trained using scikit-learn and XGBoost tools. MLR is trained simply to show the starting point. Random Forest is fine-tuned by searching through many settings to find the best tree depth and leaf size. XGBoost is tuned using a method called Bayesian optimization because it is faster for large data (Zheng, 2025). It has been observed that this method works well when there are many choices for the settings.

3.5.3 Rare-Segment Intervention

Cars that are not common are those that make up less than two percent of the data. Two ways of helping these rare cars are tested against a normal base. First, the model is told to care more about errors on rare cars through weight settings. Second, a trick called SMOTE-NC is used to create new data for rare cars for the training phase. Many people believe that this makes the machine learning-based price prediction model for second-hand vehicles fairer.

3.6 Data Analysis Methods

3.6.1 Feature Importance Analysis

Answering the first goal of the research involves looking at how features link to the price. Two ways are used for this task. First, math tests are used to see how numbers move together. Second, the Random Forest model is asked to show which features are the most helpful for the prediction. SHAP values are also calculated to see how each detail adds to the price (Lundberg & Lee, 2017). Experts claim that this way of explaining the model is needed for the Consumer Protection Act (2012, s. 3(4)(d)–(e)) and the National Automotive Policy (Republic of Kenya, 2022).

3.6.2 Model Comparison

Evaluating the performance of the three models uses RMSE and MAPE on the test data. MAPE is the most important number because it shows the error as a percentage, which is easy to understand for different car prices (Gao, 2024). RMSE is also checked because it punishes big mistakes very hard, especially for rare cars (Zheng, 2025). A math test is used to see if the differences between models are real or just luck (Bergmann & Feuerriegel, 2025). It has been observed that comparing models this way is more scientific.

3.6.3 Rare-Segment Disaggregated Evaluation

Meeting the third goal requires looking at common and rare cars separately. A car is called rare if it has very few examples in the training set. If the error for these rare cars is too high, the special fixes like SMOTE-NC are used to pick the final model. These separate numbers are shown in all the tables of the results.

3.7 Validation and Evaluation

Testing the model happens by using five-fold cross-validation on the training set. The final test is done only once on the secret test set to see how it works in the real world. Many people believe that a MAPE below fifteen percent is good enough for a tool that helps people buy cars (Mordor Intelligence, 2026). The way the model ranks features is compared to what experts in Kenya already know about cars, such as the importance of mileage and age (Onserio & Mukudi, 2025). If the model chooses very strange things, the data will be checked for errors before it is used.

3.8 Ethical Considerations

Protecting the privacy of the people who sell cars is a main goal of this scientific exploration. Only the details that everyone can see on the web are taken from Jiji.co.ke and Cheki Kenya. No names or phone numbers are saved or used by the team. The rules of the websites are followed by not going too fast and not breaking into private areas. Permission is asked from the university and the National Commission for Science, Technology and Innovation (NACOSTI) because Kenyan law requires it. Experts claim that a machine learning-based price prediction model for second-hand vehicles could be unfair if it works poorly for the cars used by people with low income. This work looks at that risk and fixes it so that the results are fair for everyone.

3.9 Chapter Summary

Kenyan context is the focus for this machine learning-based price prediction model for second-hand vehicles. Data is scraped from online selling places like Jiji.co.ke and Cheki Kenya. The problem is treated as a supervised regression task because prices are continuous numbers. Experts claim that seller wishes are different from final prices, which is a limit of this work that is mentioned clearly. The steps for cleaning and testing the models follow the path of Bergmann and Feuerriegel (2025). Three models are compared so that the best one is found for common and rare cars alike. All rules of ethics and laws are respected to make this a strong framework for the Kenyan market.

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APPENDICES

Appendix A: Budget and Work Plan

A1. Research Budget

The budget below covers all direct costs anticipated over the four months.

Table 1: Estimated Research Budget

Item	Description	Estimated Cost (KES)
IRB ethical approval	Research internal review board	3000
Internet	Data collection and online access	0
Web scraping tools	Using free software	0
Data Storage	Personal device	0

A2. Work plan

The work plan below distributes all study activities across the four months. Each cells indicate the weeks during which each activity is active.

Table 2: Work Plan for the four-Month Research Period

Activity	Month 1	Month 2	Month 3	Month 4
1. Proposal finalization and ethics submission	Weeks 1–2			
2. NACOSTI permit and IRB clearance	Weeks 1–3			
3. Web scraping (Jiji.co.ke and Cheki Kenya)		Weeks 3–6		
4. Data cleaning and deduplication		Weeks 5–7		
5. Kenya-specific feature engineering			Weeks 7–9	
6. MLR, Random Forest and XGBoost model training			Weeks 8–11	
7. Hyperparameter tuning and cross-validation			Weeks 9–12	
8. Results interpretation and SHAP analysis			Weeks 11–13	
9. Report writing and editing			Weeks 12–16	Weeks 12–16